

SENG1050

Lab 5

In this lab, you will gain experience in advance HTML topics, such as forms and tables, and advance CSS.

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Task 1 - HTML Forms

In this task, you are going to use HTML form elements and attributes to create a simple form for submitting information to a web server. Note that you need not know how the CGI script is working. The main aim of this task is for you to become familiar with the various HTML form elements, to help you create forms for the assignments.

1. Make a new copy of your `template.html` file from Lab 2 and open it in your favourite plain-text editor (e.g., notepad or textpad).
2. After adding an appropriate title and level-1 heading, insert form tags into the body of your document:

```
<form>
</form>
```

3. In the opening of the form tag, include the attributes `method="post"` and `action="https://jkorpela.fi/cgi-bin/echo.cgi"`.
4. Within the form, place elements to accept the following information:
 - The applicant's first name (`<input type="text">`, include the `placeholder="First name only"` and `required` attributes)
 - The applicant's last name (similar to first name)
 - The applicant's date of birth (`<input type="date">`, include the `required` attribute)
 - The applicant's age (`<input type="number">`, include the `min`, `max`, and `value` attributes to set the minimum, maximum, and default ages)
 - The applicant's email (`<input type="email">`, include the `placeholder` and `required` attributes)
 - The applicant's website (`<input type="url">`, include the `placeholder` attribute)
 - The applicant's phone number (`<input type="tel">`, include the `placeholder` and `required` attributes)
 - The applicant's skill level (`<input type="range">`, include the `min="1"`, `max="10"`, and `value="1"` attributes)
 - The applicant's address (`<textarea>`, include the `rows="5"` and `required` attributes)
 - The applicant's postcode (`<input type="text">`, include the `size="4"` and `maxlength="4"` attributes)
 - The applicant's state (`<select>` and `<option>`. Use one `<option>` tag for each of the states (NSW, QLD, SA, TAS, VIC, WA))
 - The applicant's card type (create three `<input type="radio">` buttons; include the `name` attribute to ensure that only one of the three card types (visa, mastercard, amex) can be chose; set visa to be the default choice using the `checked` attribute)
 - The name on the card (similar to first name/last name)
 - The card number (`<input type="number">`)
 - The card's security code (`<input type="number">`, include the `min="001"` and `max="999"` attributes to enforce the security code to be a 3 digit positive number)
 - The card's expiry date (`<input type="month">`)

- For each form control, associate a label with the `<label>` tag (use explicit labelling).
- Group the form controls into four groups with `<fieldset>` and `<legend>` tags as shown in the following figure.
- There should also be a button to submit the form, and a button to reset the form.

Your form could look something like this:

Register for HTML5 Geeks Club

Personal details

First Name Last Name

Date of Birth Age

Email

Personal Website

Telephone

HTML5 Skill level (1 Newbie - 10 Expert)

Billing address

Address

Post code State

Card details

Card type

☒ VISA ☐ Mastercard ☐ AMEX

Name on Card

Card number

Security code

Expiry Date

After submitting the form, you should see a list of the information that the server-side CGI script received from your web browser.

Task 2 - HTML Tables

In this task, you are going to use an HTML table to display some statistical data from the web.

1. Make a new copy of your `template.html` file from Lab 2 and open it in your favourite plain-text editor.
2. After adding an appropriate title and level-1 heading, insert table tags into the body of your document:

```
<table>
</table>
```
3. Inside the table tags, insert the `<thead>`, `<tbody>` and `<tfoot>` tags (they **must** be in that order and don't forget each corresponding closing tags).
4. Access the [Distribution of Top-Level Domain Names by Host Count, January 2013](#) page hosted by the [Internet Systems Consortium](#). You will use the data from this site (just the first 6-7 rows will do) to create your table.
5. In the head (`<thead>`) of your table , create a row (`<tr>`) for the headings of the table. Populate each cell (column) of this row using the table heading (`<th>`) tag. You will use the content from row 1 of the [original data](#).
6. In the body (`<tbody>`) of your table , create the necessary rows (`<tr>`) for the data of the table. Populate each cell (column) of these rows using the table data (`<td>`) tag. You will use the content from row 3 to 7 of the [original data](#).
7. In the foot (`<tfoot>`) of your table , create a row (`<tr>`) for the foot of the table. Populate each cell (column) of this row using the table data (`<td>`) tag. You will use the content from row 2 of the [original data](#).
8. Finally, add an appropriate `<caption>` to the table, which includes a hyperlink to the source of the [original data](#).

Now, you can add visual formatting suggestions for your table using embedded CSS styles.

1. In the content of the `<head>` tag, place a CSS `<style>` tag:

```
<style>
</style>
```
2. Inside this `<style>` tag, add CSS rules to suggest visual formatting for your tables. Try to replicate the styling of the table in the figure below:
 - You will have to add `class` attributes to some tags to differentiate certain rows and columns.
 - You should at least change the text alignment of the numbers to be right aligned, without changing the alignment of the rest of the table's cells (hint: apply `text-align: right` to the body and footer of the table. Then, override the cells in the first column of the table and the last column of the body of the table with `text-align: left`).
 - You should have different background colours for the cells of the head, body, and foot of the table.

Don't forget to validate your [HTML](#) and [CSS](#).

Your completed table should look something like this (but with current figures and only a few rows of data).

Internet Statistics

Distribution of Top-Level Domain Names by Host Count Jan 2003 (ISC)

Domain	Hosts =	All Hosts -	Dup Hosts	Level 2 Domains	Level 3 Domains	
net	61945611	64522261	2576650	181746	12354344	Networks
com	40555072	84169424	43614352	1716051	12594151	Commercial
jp	9260117	9377758	117641	13889	141443	Japan
edu	7459219	7688934	229715	3994	2533055	Educational
arpa	6387463	23401589	17014126	18	3817	Mistakes
it	3864315	3905353	41038	31831	316245	Italy
ca	2993982	3397065	403083	17007	974715	Canada
de	2891407	2996705	105298	138718	778042	Germany
uk	2583753	3192125	608372	66	57921	United Kingdom
au	2564339	2627795	63456	45	43625	Australia
TOTAL	171638297	238188683	66550386	2659678	35982449	

More tasks with table (optional/homework)

Change your table with 'rowspan' and 'colspan' attributes.

- Using 'colspan' property, merge cells (r3,c5) and (r3,c6) where (rX,cY) represent the cell at row X, column Y of the body of the table.
- Using 'rowspan' property, merge cells (r5,c2), (r6,c2) and (r7, c2).
- Insert random numbers in these merged cells.

Task 3 - Advance CSS Layouts

In this task, you are going to format a web page using the advanced CSS layout properties. These will be implemented as embedded CSS styles.

1. Download the template files provided through task3.zip in the Week 6 Canvas module.
2. The first step is to format the navigation links into a horizontal navbar by doing the following.
 - Float each list item left and remove the bullets.
 - Place a background colour on the parent element. You'll need to set `overflow: hidden` so that the parent expands around the floated elements.
 - Now, change the background colour of individual list items on hover.
 - Notice the difference between the list item and anchor elements, we want the entire box to be a clickable link. Remove margins and padding on the unordered list element and place them on the anchor tags instead. You'll need to set the display property to either block or inline-block for this to take effect.
 - Now, we want to make the navbar stick to the top of the screen as the user scrolls down. Set the position property of the navbar to sticky. Why doesn't this work? To fix this problem, set the display property of the <header> element to 'contents'.
 - The end result could look something like the following image:

Featured Books:

3. Now, it's time to handle that huge block of sample text:

- Set a fixed height for the text-block of about 30em.
- Set the overflow property to scroll.

4. Now, let's look at the featured books section:

- We'll use a columnar layout for this, set the columns property to produce 3 columns.
- Shrink the images, and have them float left so that the text wraps around on the right side.
- To prevent an article element from spanning multiple columns, target the article element with the CSS rules: `break-inside: avoid; clear: left; overflow: hidden`. You may also set a bottom margin to your liking.
- The end result could look something like this:



5. Finally, we will use a flexbox to layout the gallery section:

- Set the gallery to be a flexbox with a fixed height of about 10cm.
- Enable scrolling using the CSS rule `overflow-x: scroll`.
- Format the images to be the same height using `height: 100%` and put some padding between them.

6. Can you make this look better? Try different properties, cleanup some of the rough edges of the layout, and make the page look as good as you can.

Be sure to [validate](#) your HTML page.